Hamromaster
सुनौलो भविष्य हातमा **Hamromaster** को साथमा

Hamromaster
COMPLETE NOTES

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For Hamromaster Online Classes

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Hypothesis Testing

Day-1

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Testing of hypothesis

- **Non-parametric test**
  - χ^2-test
    - 1 fixed question [10 marks]
- **Parametric test.**
  - * Z-test → 1 fixed question
  - * t-test → 1 fixed question
  - * F-test

1 Long	→ 10 marks
1 short	→ 5 marks
Maximum	→ 25 marks
Minimum	→ 20 marks.

Hypothesis ⇒ Assumption ⇒ Population [May be True or False]

⇓

Validity Test

⇓

Sample द्वारा

Finite popn

Infinite popn.

Entire process = Testing of Hypothesis.

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Def^n of Hypothesis Testing

* An assumption about the population parameter is called hypothesis. It may be true or false. So, its validity of population parameter on the basis of sample evidence is called testing of hypothesis.

It means, after a hypothesis testing procedure we came to know whether our a assumption was true or false.

Types of Hypothesis:

(No significance difference)

a) Null Hypothesis (H_0): · It is the hypothesis of no difference.
· It tells that the assumption is true.
· Denoted by H_0.

b) Alternative Hypothesis: · It is the hypothesis of difference.
· It tells that the assumption is false.
· Denoted by H_1 or H_A.
· It is complementry of Null hypothesis.

Level of Significance (α) / Size of critical region:

* The maximum size of rejecting the null hypothesis is called level of Significance.
* The level of risk for rejecting true null hypothesis.
* The probability of type I error is called level of significance. $\alpha = P(\text{Type I error})$
* It is denoted by 'α'.
* Under this level, we try to obtain critical value from table

प्रश्नले α नदिएमा हामीले जहिले पनि $\alpha = 5\%$ मान्नु पर्छ। कहिलेकाही Confidence level भनेर, जस्तो कि 95% $\boxed{\alpha = 100\% - 95\% = 5\%}$

Power of test

* The probability of rejecting H_0 when it is false.

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$$\text{Power of test} = P(\text{reject } H_0 \text{ when it is false})$$
$$= 1-\beta$$

* It is also known as prob. of correct decision.

V.V Imp

Types of error in Hypothesis Testing

* Type I error: The rejection of (H_0) null hypothesis when it is true is called type I error.

* Type II error: The acceptance of (H_0) null hypothesis when it is false is called type II error.
  - The prob. of making type II error is denoted by β.

$$\therefore \beta = P(\text{Type II errors})$$

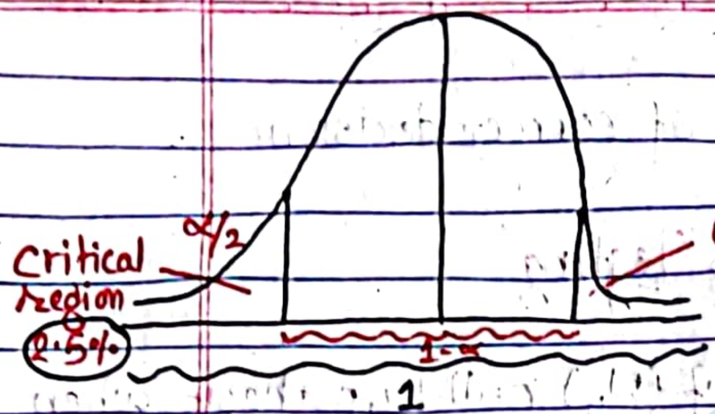
- धुम्रपान स्वास्थको लागि हानिकारक छ..... [True statement]
  Accept गरेर नरवाए ... [Error हुन नथियो]
  Reject गरे [Error हुन्छ।]

Critical Region	Acceptance Region.
The region of the sample space where the null hypothesis is rejected when it is true is called critical region. Also known as rejection region which is denoted by W.	The mutually exclusive and complementry region to the critical region under sample space is called acceptance region. It is denoted by $\bar{w}$

Note: Sample Space. All possible outcome obtained from all possible sample constitute a space called sample space. It is denoted by Ω or S.

Day-2

One Tailed and Two Tailed Tests



$$1-\frac{\alpha}{2}-\frac{\alpha}{2}$$

Let $\alpha = 5\%$

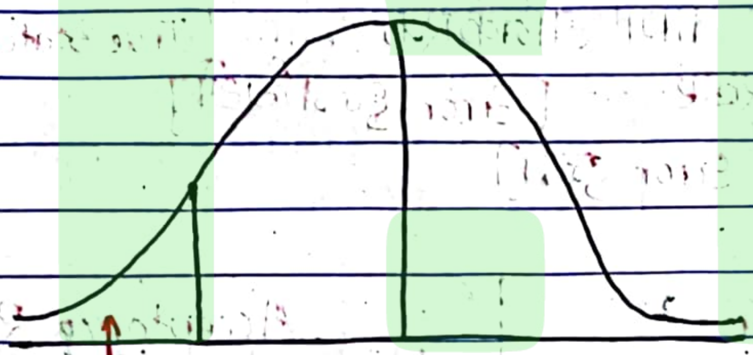
$$= 1-\left[\alpha/_2+\alpha/_2\right]$$

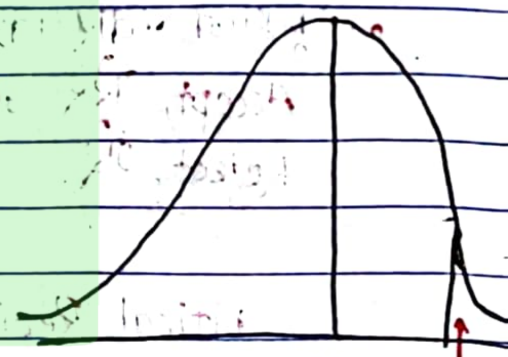
$$= 1-\left[\frac{\cancel{2}\alpha}{\cancel{2}}\right]$$

$$= 1-\alpha$$

$1-\alpha$ = Acceptance region or critical region.

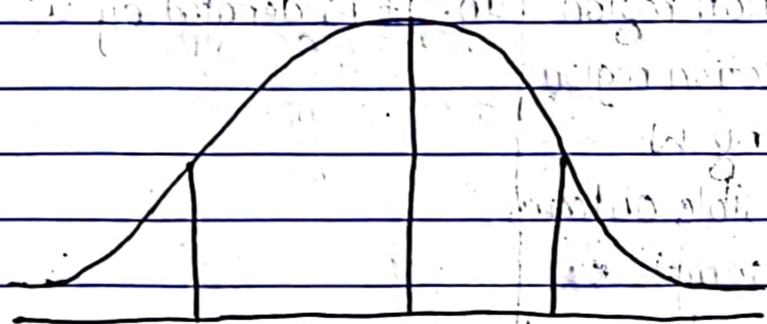
One tail:





Left tail test $\xrightarrow{\text{One tail}}$ Right tail test

Two tail:



Two tail test

Direction of difference → specified → one tail
→ not specified → two tail

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* Ram income is greater than Hari → One tail
→ specified

Low | less than | Taller | decrease | Smaller | greater | more than | improvement | under | higher | majority | minority | superior | inferior | At least | At most (within shorter success effective only

* Ram income is differ from Hari → Two tail

[illegible]

Z-test
T-test
F-test
} Parametric Test [10-15 marks] Non-Parametric Test { K^2- test
(5-6) question

Z-Test [Test of Large Sample Size]

* प्रश्न चिन्ने आधारहरु:

1. प्रश्नमा दिएको sample size 30 वा 30 भन्दा बढि भएमा त्यो प्रश्न Z-test भनेर बुझ्नु पर्छ।

If the sample size are greater than or equal to 30 then, that is called large sample size. [Z-test]

2. यदि प्रश्नले हामीलाई (population standard deviation) दिएको छ तर sample size 30 भन्दा कम भएता पनि त्यो प्रश्न Z test नै हो भनी बुझ्नु पर्छ।

Types of Z-test

- Mean Test
  - Single mean (1 sample size)
  - Double mean (2 sample size)
- Proportion Test
  - Single proportion (1 sample size)
  - Double proportion (2 sample size)

* यदि प्रश्नमा S.D दिएको छ भने त्यो प्रश्न Mean को हो, नत्र proportion को हो भनेर बुझ्नु पर्छ।

Single Mean :-

Sample	Population
* Sample size = n	Pop^n size = N
* Sample Mean = $\bar{x}$	Pop^n mean = μ
* Sample s.d. = σ	Pop^n s.d = s
* Sample variance = s^2	Pop^n variance = σ^2
* Sample proportion of:	Pop^n proportion of:
• success = p	• success = P
• failure = q	• failure = Q
$p+q=1$	$P+Q=1$

Z-test for single mean :

* It is the comparative study bet^n sample mean and pop^n mean.

- one sample size ($n \geq 30$)
- one sample mean ($\bar{x}$)
- one pop^n mean (μ)
- one sample s.d or one pop^n s.d.

प्रश्नको समाधान गर्ने तरिका :-

Step I: Null Hypothesis (H_0) : $\mu = \mu_0$ (i.e. There is no significance difference betⁿ sample mean and popⁿ mean)

OR

The popⁿ mean has specified value of μ_0

Step II: Alternative Hypothesis (H_1) : $\mu_0 \neq \mu_0$ (Two tailed test)

[There is significance betⁿ sample mean and popⁿ mean]

OR

The popⁿ mean hasnot specified value of μ_0

one tail.
- $H_1 : \mu > \mu_0$ यदि प्रश्नले (right tail) छ भने (चुच्चो right तिर)
- $H_1 : \mu < \mu_0$ (left tail) (चुच्चो left तिर)
  - i.e. [the popⁿ mean is greater than sample mean]
  - i.e. [the popⁿ mean is less than sample mean]

Step III: Test statistics: Under H_0

$$Z_{cal} = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \text{ or } \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

→ Standard error (S.E)

Step IV: Level of significance (α) or Size of Critical region.

Step V: We find critical value or tabulated value with the help of significance (α).

Step VI: Comparision and decision:

① Since, $|Z_{cal}| > |Z_{tab}|$ Hence it is significant. H_0 is rejected and H_1 is accepted. So, we conclude that

जसलाई accept गरेको छ त्यसैलाई भाषा सार्ने।

② $|Z_{cal}| < |Z_{tab}|$ Hence it is not significant. H_1 is ~~accepted~~ rejected and H_0 is accepted. So, we conclude that

जसलाई accept गरेको छ त्यसलाई भाषा सार्ने

Non-Parametric Test

Chi-sq. test (χ^2) [10 marks]

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Parametric Test: Parametric Test is a statistical test which makes certain assumption about the distribution of unknown parameter of interest and Thus test statistic is valid under these assumption.

Non-Parametric Test: Non-parametric Test are defined as those statistical test of hypothesis in which no parameter is involved and which are based on ~~under~~ or der statistics.

Chi-sq. test (χ^2)

प्रश्न चिन्ने आधार

→ Just frequency मात्र हुन्छ भन्नाले प्रश्नमा Mean, s.d, जस्तो information हुँदैनन् ।

Formula: $\boxed{\chi^2 = \sum \frac{(O-E)^2}{E}}$

where, O = observed frequency
E = expected frequency.

Note: Observed frequency जहिले पनि प्रश्नमा हुन्छ भने expected frequency जहिले पनि हामी आफैले calculate गर्नु पर्छ ।

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Assumption/properties of χ^2-test.

(i) The total observed freq. should be reasonably large (i.e. $N \geq 50$)

(ii) Total observed freq. and total expected freq should be equal
i.e. $\Sigma O = \Sigma E = N$
↳ grand total.

(iii) हामीले findout गरेको expected freq. यदि 5 भन्दा कम आएमा त्यसलाई pooling गरेर 5 भन्दा बढि बनाउनुपर्छ।

Note: सामान्यतया:
Degree of freedom (d.f.) = (n-1)
तर: यदि Expected frequency 5 भन्दा कम आउदा हामी pooling गरेका छुन्छौं भने त्यस्तो बेला
degree of freedom (d.f.) = $n-1-k_1-k_2$..

6
6
1 → $1 = k_1$
13
14

$k_2 = 1$ otherwise 0.

Binomial distribution	(n, p) parameter
Poission distribution	λ.

V.V.I. χ^2-test for independent of attributes.

Independent
→ छ: There is no relation (H_0)
→ छैन There is relation (H_1)

No association	Association
↓	↓
Relation	Relation

सबैलाई समान व्यवहार → No.
diff – Association.

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प्रश्न कस्तो हुन्छ त?

- Is there any significant relation or association relation betⁿ row and column.
- Is row factors depends upon column factor? or vice versa.
- Is row factor independent or vary from column factor? or viceversa

प्रश्नको समाधान गर्ने तरिका!

Step 1: Setting Hypothesis:

Null Hypothesis (H_0): There is no significant relation bet^n row factor and column factors.

Alternative Hypothesis (H_1): There is significant relation bet^n row factor and column factors.

Step 2: Test statistics under H_0,

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

where, O = Observed freq.
E = Expected freq.

Note: $E = \frac{RT \times CT}{N}$

where, RT = Row Total →
CT = Column Total ↑
N = Grand Total

Step 3: Level of significance α%

Step 4: Degree of freedom (d.f) = $(c-1)(R-1) - k_2$

where, k_2 = जतिवटा poll गरेको त्यती राख्ने

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Step 5: Critical value or tabulated value with the help of $\alpha\%$ and d.f.

Step 6: Comparison and decision.

$|Z_{cal}| < |Z_{tab}|$. It is significant. H_0 is accepted and H_1 is rejected. We conclude that ---
---

$|Z_{cal}| > |Z_{tab}|$ It is not significant. H_0 is rejected and H_1 is accepted. We conclude:

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[10marks] T-test [Fix]

↓

Small Sample Test.

✻ Sample Size 30 भन्दा कम भएमा त्यो प्रश्न T-test हो भनेर बुझ्ने तर कुनै बेला हामीलाई प्रश्नले pop^n s.d. दिएमा Sample size 30 भन्दा कम भएता पनि त्यो प्रश्न फेरी z-test हुन्छ।

pop^n s.d छ भने → z-test

pop^n s.d छैन sample size 30 भन्दा कम - T-test.

Types of T-test.

- Mean Test
  - Single Mean (5 marks)
  - Double Mean
    - Independent Mean
    - Depedent Mean [Pair T-test]
- Correlation Test

→ 10 marks

Single Mean. It is the comparative study betⁿ sample mean and population mean.

- One sample size $(n<30)$
- one sample mean $(\bar{x})$
- one pop^n mean (μ)
- one unbiased or one biased variance

अरुमा dependent नभको

unbiased → S^2 ↳ We have to calculate

biased → s^2 ↳ Question ले दिन्छ

प्रश्नको समाधान गर्ने तरिका :-

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Step 1: Setting Hypothesis...

Null Hypothesis(H_0): $\mu = \mu_0$ [i.e. There is no significance diff. betⁿ sample mean and popⁿ mean]

Alt. Hypothesis (H_1): $\mu \neq \mu_0$ [Two tailed]
[ie. There is significance diff. betⁿ sample mean and popⁿ mean

one tailed
- $H_1: \mu > \mu_0$ [right tailed]
  [i.e. Sample mean is greater than popⁿ mean]
  [popⁿ mean is greater than μ_0]
- $H_1: \mu < \mu_0$ [left tailed]
  [i.e. Sample mean is smaller than popⁿ mean]
  [popⁿ mean is less than μ_0]

Step 2: Test Statistics under H_0,

$$t_{cal} = \frac{\bar{x} - \mu}{\sqrt{\frac{S^2}{n}}} \rightarrow \text{ठूलो } S$$

unbiased

$$t_{cal} = \frac{\bar{x} - \mu}{\sqrt{\frac{s^2}{n-1}}}$$

biased

Step 3: degree of freedom (d.f) = n-1

Step 4: Level of significance (α) = _%

Step 5: Critical value or tabulated value...

Step 6: Comparison and Decision.

Since, $|t_{cal}| > |t_{tab}|$ ∴ It is significant, H_0 is rejected, H_1 is accepted. We conclude that -----

$|t_{cal}| < |t_{tab}|$ ∴ It is not significant. H_0 is accepted and H_1 is rejected. We conclude that ----

Calculation of Unbiased Sample variance. [S]

(i) $$S^2 = \frac{1}{n-1}\left[\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right]$$ [Direct formula]

(ii) $$S^2 = \frac{1}{n-1}\Sigma(x-\bar{x})^2$$ [Actual mean method]

Note: (i) Sum of observation = Σx

(ii) Sum of square of observation = Σx^2

(iii) Sum of sq. of deviation from mean = $\Sigma(x-\bar{x})^2$

Note: Below or Above → two tailed test.

More or less → two tailed test

More or not → one tailed test

Effective or not → one tailed test

Less or not → one tailed test.

A random sample of 16 values from a normal pop^n showed a mean of 41.5 and the sum of square of deviation from this mean equal to 135 sq. inch. Show that the assumption of a mean of 43.5 inch for the pop^n is not reasonable also 95% fudicial limit.

Soln:- Here,

Sample size $(n) = 16$

Sample mean $(\bar{x}) = 41.5$

Sum of sq. of deviation from mean $\Sigma(\bar{x}-x)^2 = 135$

pop^n mean $(\mu) = 43.5$

We know,

$$S^2 = \frac{1}{n-1}\Sigma(\bar{x}-x)^2$$

$$= \frac{1}{16-1} \times 135$$

$$= \frac{1}{15} \times 135$$

$$= 9$$

Step 1: Setting Hypothesis [Two tailed]

Null Hypothesis (H_0) : $\mu = 43.5$ [i.e. There is no significant diff. betn. pop^n mean and sample mean]

Alt. Hypothesis (H_1) : $\mu \neq 43.5$ [i.e. There is significant diff. bet^n pop^n mean and sample mean]

Step 2: Test statistics under H_0

$$t_{cal} = \frac{\bar{x}-\mu}{\sqrt{\frac{S^2}{n}}} = \frac{41.5-43.5}{\sqrt{\frac{9}{16}}} = \frac{-2}{\frac{3}{4}}$$

$$= -2.67$$

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